11-15

Topics covered on Test 3 are posted

HW sol'ns for 12 + 13 will be posted no later than tomorrow.

Test is Wed.

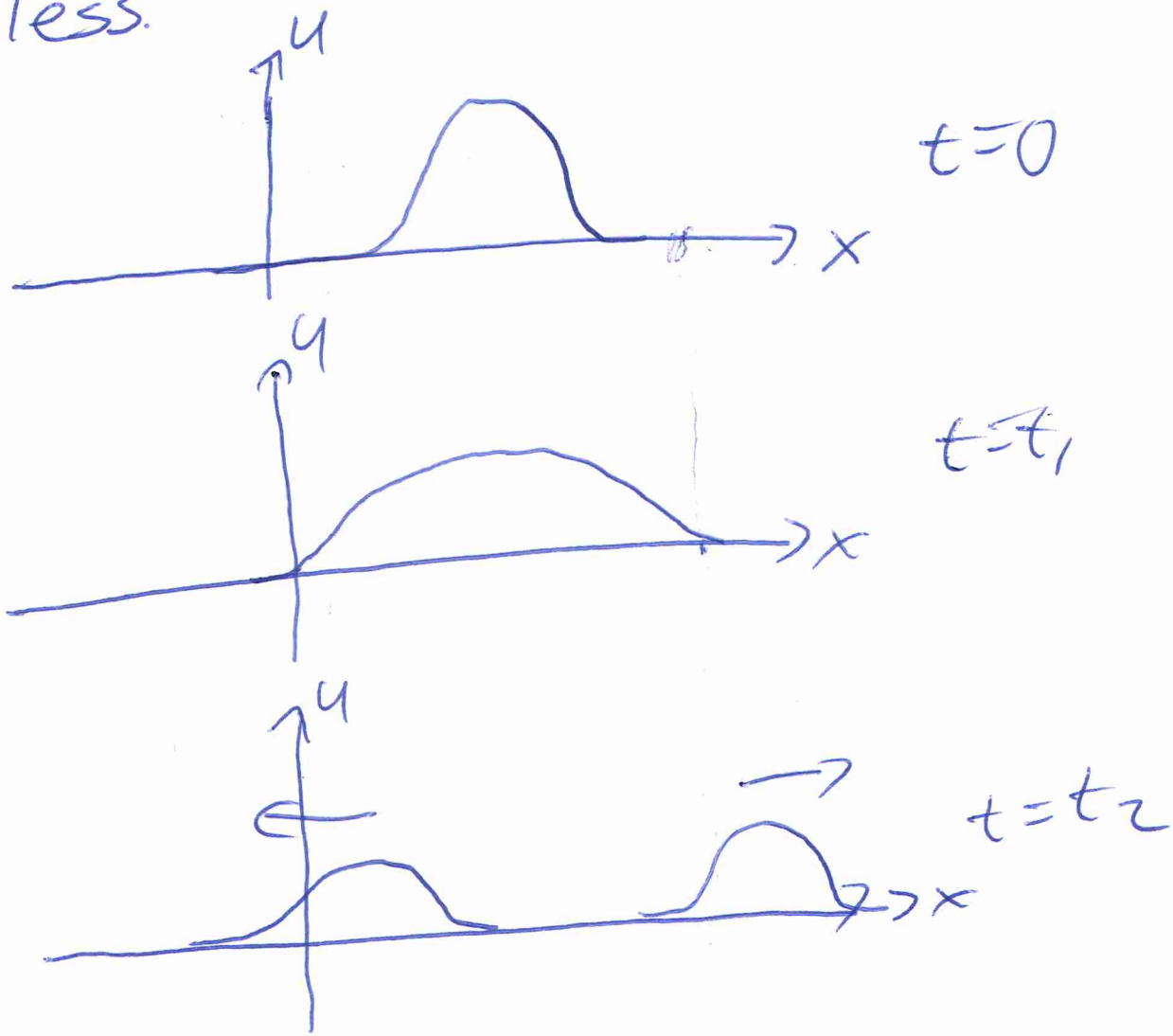
Project (540 students only) (option 440)

please see me about topic if you have not already.

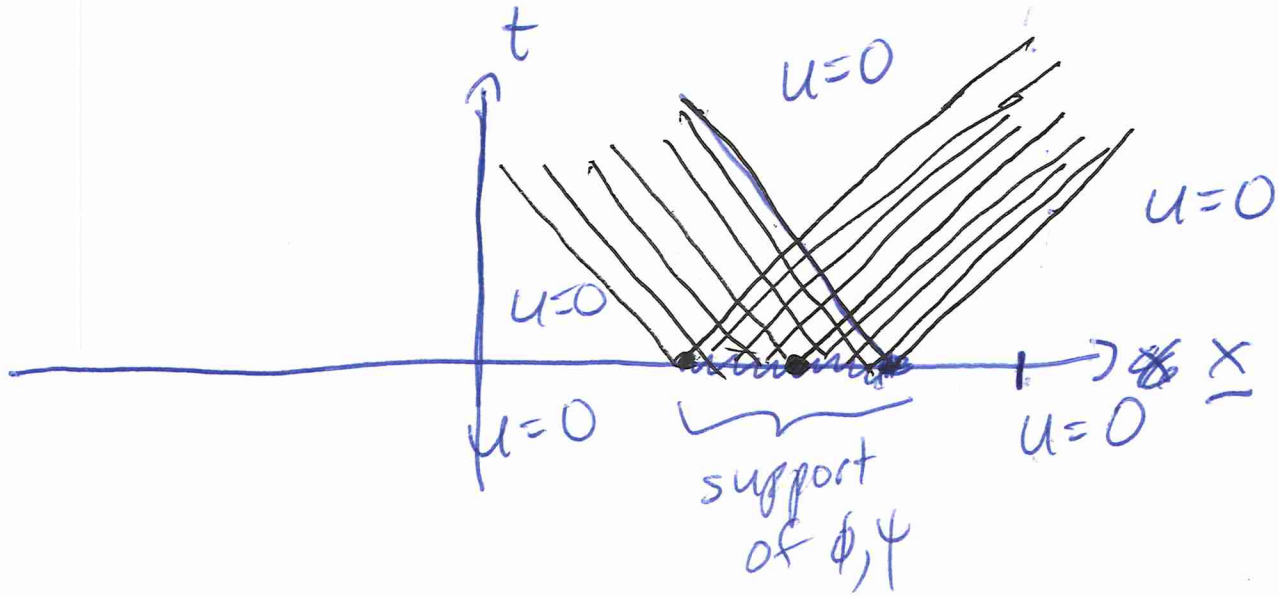
AI (chatgpt etc) policy to be posted.

Huygen's Principle: Range of influence is now less.

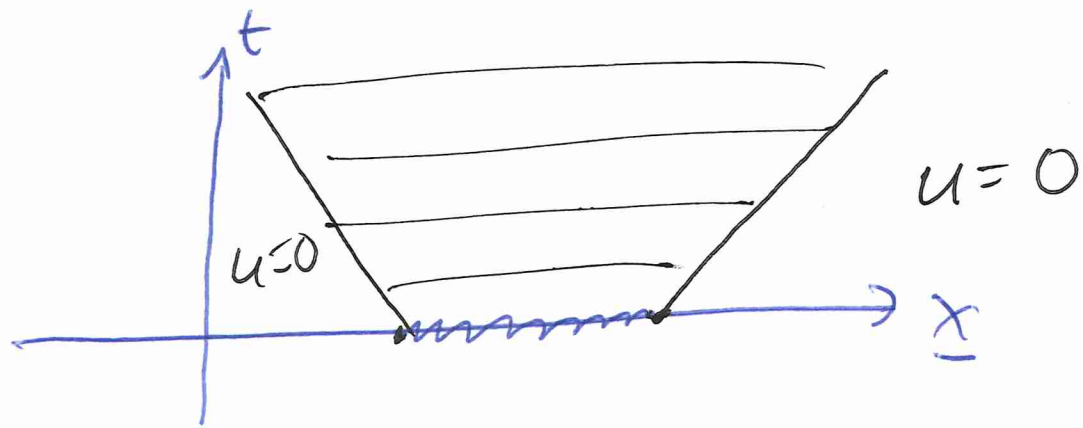
$n=1$,
$\psi=0$
displacement
is ~~like~~ behaves
as it does for
$n=3,5,\ldots$



Range of influence for $n = 3, 5, 7, \ldots$



Range of influence for $n = 2, \ldots$



<u>Note</u> For all Thms thus far in ch 8, we have assumed (at least) that $\phi \in C^2(\mathbb{R})$ and $\psi \in C^1(\mathbb{R})$ so that $u \in C^2$.

What if ϕ (or ψ) is only piecewise continuous?

Then u would only be sectionally continuous & how would we make sense of $u_{xx} = u_{tt}$?

This gets us to "generalized sol'n" involving "distributions" (generalization of f'n)

ex) $\delta(x)$ $\quad \frac{d}{dx}[H(x)] = \delta(x)$

But from a practical stand pt, we can just continue using our formulas.

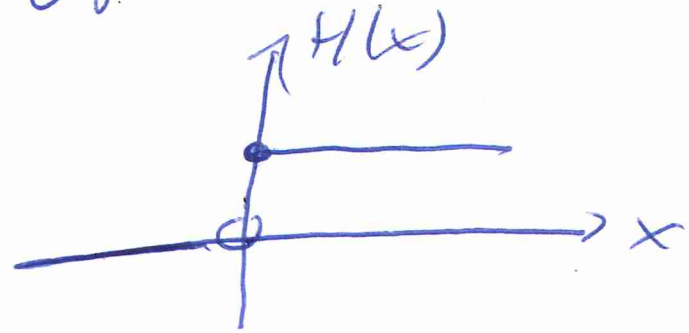
We will see an example in 2 more pp.

Discontinuous IC?

ex) $n=1$ $\quad u_{xx} - u_{tt} = 0$ $\quad$ IVP

Behave like $n=3$

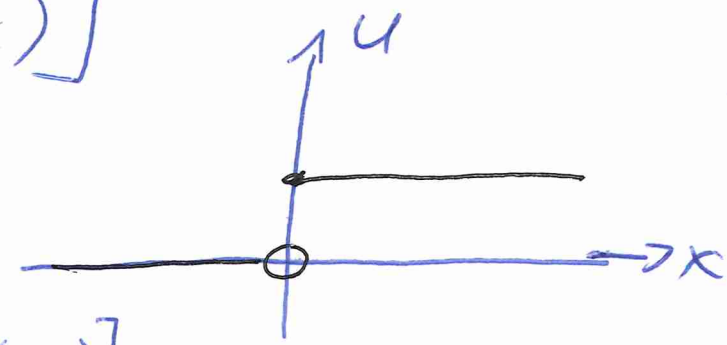
$$\begin{cases} \phi(x) = H(x) = \begin{cases} 0 & x<0 \\ 1 & x \geq 0 \end{cases} \\ \psi(x) = 0 \end{cases}$$



D'Alembert's sol'n:

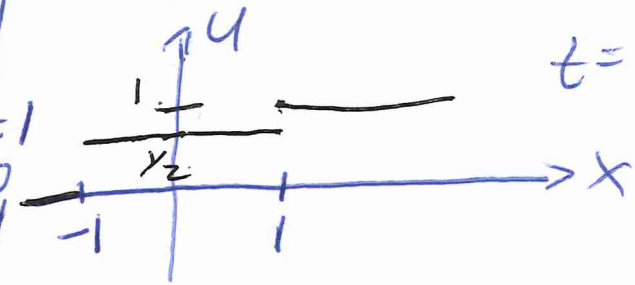
$$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \cancel{\psi(x')}^{\,0}\, dx'$$

$$= \frac{1}{2}\left[H(x+t) + H(x-t)\right]$$

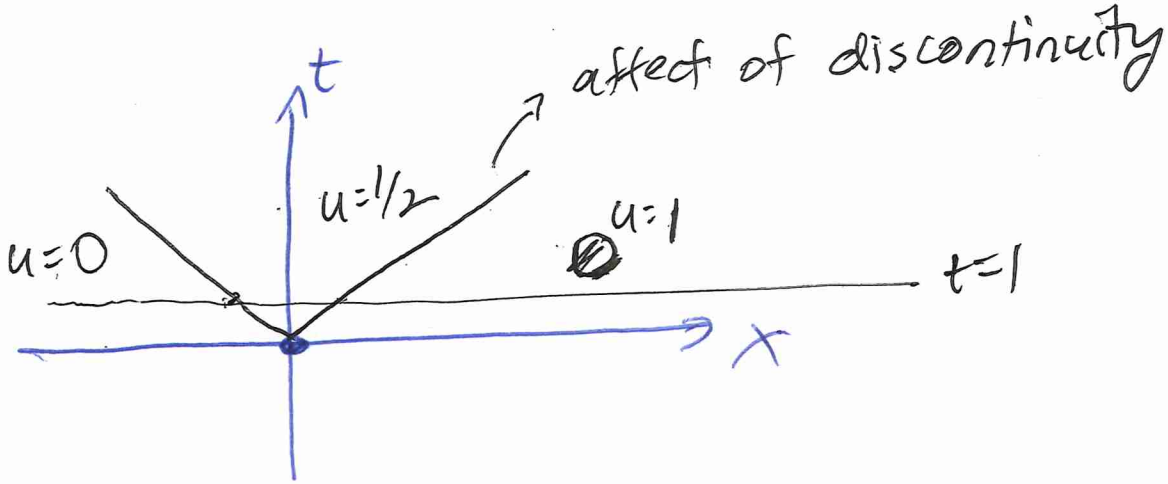


$t=0$

$t=1$? $\quad u(x,1) = \frac{1}{2}\left[\underbrace{H(x+1)}_{\substack{x+1>0 \\ x>-1}} + \underbrace{H(x-1)}_{\substack{H=1 \\ x-1>0 \\ x>1}}\right]$



$t=1$


affect of discontinuity
t
u=1/2
u=0
u=1
t=1
x

Semi-infinit string (Ch 8 §7)

$n=1$

$$u_{xx} - u_{tt} = 0 \qquad x>0,\ t>0$$

$$\left.\begin{aligned} u(x,0) &= \phi(x) \\ u_t(x,0) &= \psi(x) \end{aligned}\right\} \quad 0<x<\infty$$

$$u(0,t) = 0 \qquad t>0$$

<u>Aside</u>

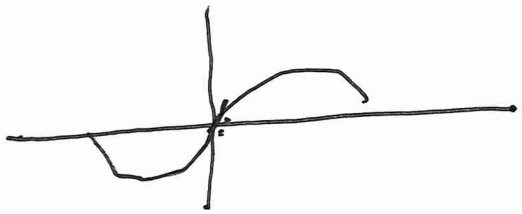
$\Delta u = 0$ in Ω

$u = f(x)$ on $\partial\Omega$

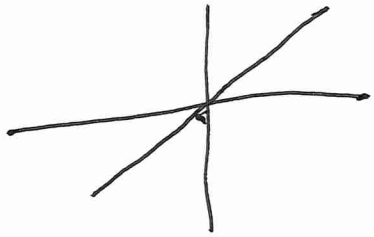
To <u>solve</u>

Extend data (ϕ, ψ) to the left of $x=0$ in an odd way & then solve as ~~a~~ an IVP.

(d'Alembert's sol'n or Fourier series)



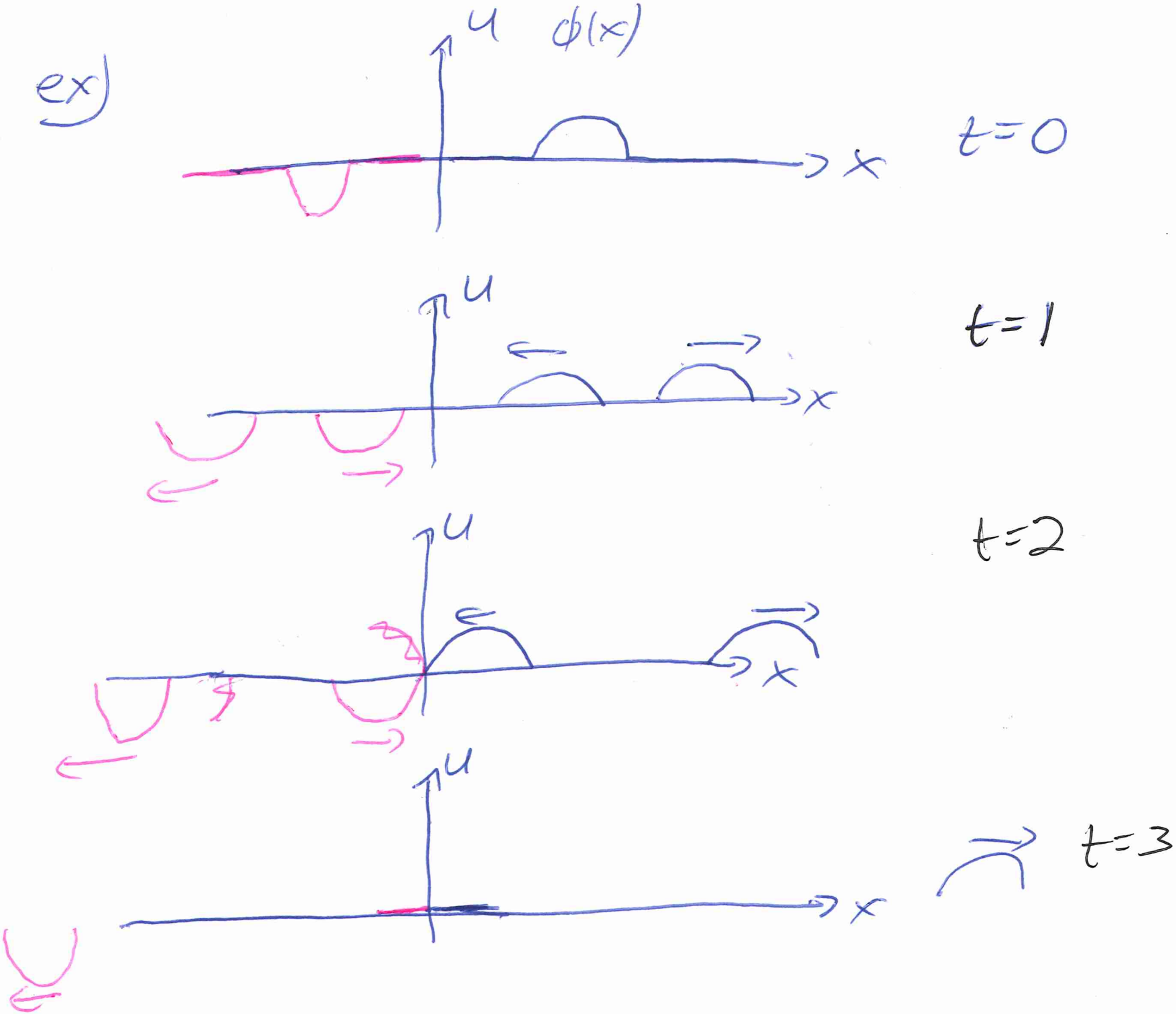
$\sin(x)$



x

~~A~~ all continuous odd fns are zero at $x=0$.

ex)



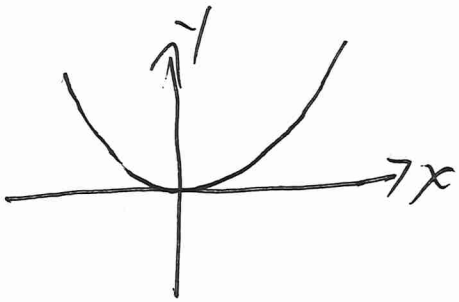
$t=4$

For Neumann B.C.

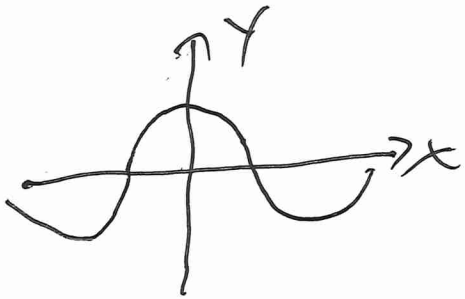
$u_x(0,t)=0$

extend (ϕ,ψ) evenly

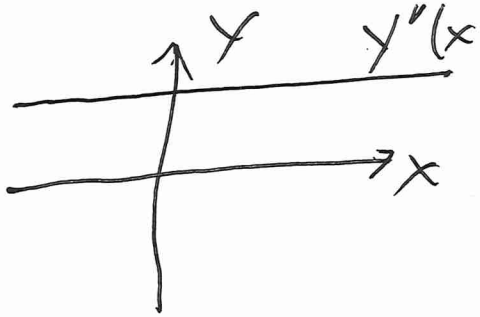
All ~~con~~ C^1 even f'ns satisfy $y''(x=0)=0$



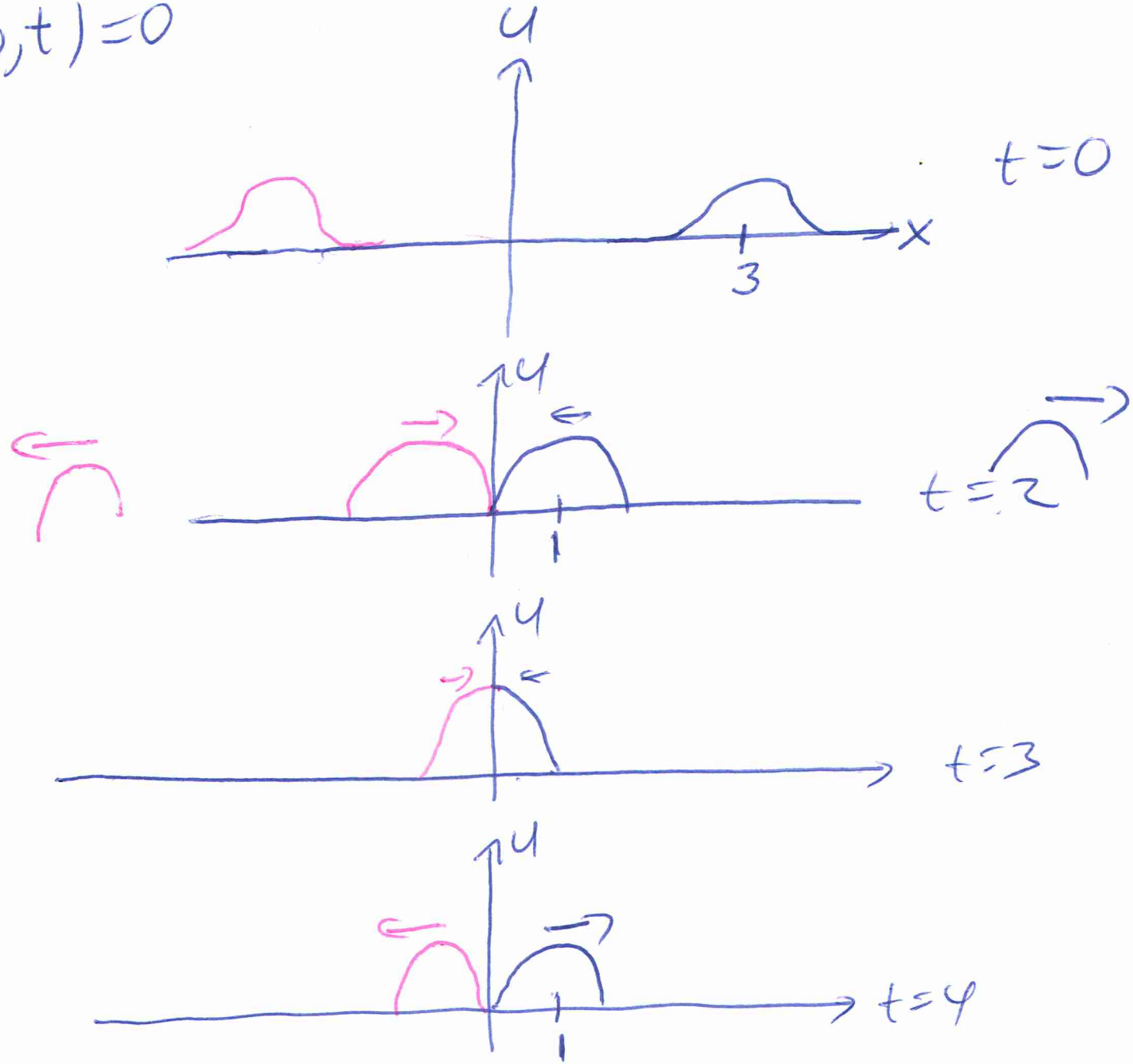
x^2



$\cos x$



$y=5$

Neumann cond'n

$u_x(0,t)=0$